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Ecology of sprouting in woody plants: the persistence niche

William J. Bond and Jeremy J. Midgley

Many woody plants can resprout and many ecosystems are dominated by resprouters. They persist *in situ* through disturbance events such as fire, flooding or wind storms. However, the importance of ‘persistence’ in plant demography has been neglected in favour of ‘recruitment’. Thus much research on plant regeneration, conservation and evolution has focused on the importance of safe sites, seed and seedling banks, dispersal and germination with the implied importance of *de novo* replacement rather than persistence. Recent research shows a growing appreciation for the role of sprouting as a form of persistence in a diversity of ecosystems and tradeoffs between the two regeneration modes.

In a seminal paper in 1977, Peter Grubb¹ introduced the concept of the ‘regeneration niche’ to help explain the coexistence of similar species. Grubb noted that species that share much the same life form, phenology and habitat range might nevertheless have different seedling requirements. For example, ‘when a whole large tree is blown over’ the large gap thus formed would favour light-demanding seedlings, whereas smaller gaps would favour shade-tolerant recruits. The literature on the ecology of seeds and seedlings has expanded enormously since this time. Seed ecology is widely studied because of its assumed importance in the population growth of plants. Seed traits are included in general systems for classifying plant functional types^{2,3}. The same broad recognition has not been given to the mode of persistence of established plants. When a tree is blown over, gaps might not be filled by seedlings but by shoots sprouting from the fallen tree. Sprouts grow much faster than seedlings and can quickly reoccupy their own gaps. Sprouting ability can have major impacts on plant populations: turnover of populations is reduced; the effects of disturbance are minimized; and dependence on seeds for population maintenance might become negligible. Species differ in their sprouting ability, and both strong and weakly sprouting species occur in diverse ecosystems^{4,5}. Here, we review studies of the phenomenon emphasizing emerging generalizations and implications for the population biology of woody plants.

The ecology of sprouting has analogies to clonality⁶. However, although clonal plants generally sprout, only a small fraction of woody sprouters are clonal and capable of vegetative spread. Sprouting response is difficult to quantify partly because there is a continuum of responses to disturbances of varying severity⁷. This could partly account for its absence from general plant strategy schemes (Box 1). After the least severe disturbance, such as damage caused by caterpillar feeding, plants replace lost tissues by sprouting from buds. As the severity of disturbance increases, plants diverge in their ability to recover by sprouting (Fig. 1). Most interest has been in disturbances that can potentially kill plants such as fire, hurricane damage, drought, flooding, herbivory and landslides, as well as anthropogenic disturbance such as forest clearing. In Mediterranean type shrublands, where crown fires are relatively frequent, shrub species are often either killed outright by burning (nonsprouters or ‘seeders’) or recover vegetatively from roots or stems (sprouters)^{4,8}. Classifying sprouting behaviour is harder in forests and woodlands because species often diverge in their sprouting response at different life history stages^{5,7,9}. Some species never sprout; in some species, sprouting ability increases with size to reach a maximum in adult stages, although in other species sprouting is common in juveniles but adults are unable to sprout (Fig. 2). Some forest trees retain a bud bank, sprouting continuously with or without disturbance and thus come close to immortality. Examples include *Tilia cordata* (small-leaved lime) in Britain, whose northern populations have survived by sprouting since climates cooled and reproduction ceased 5000 years ago¹⁰. *Ginkgo biloba* (maidenhair tree) might owe its survival in China to the extraordinary persistence of trees that resprout from specialized basal swellings on the trunk¹¹.

Sprouting is common, and might be the ancestral state, in woody angiosperms¹². Most conifers do not sprout although there are exceptions in several unrelated lineages, including species of *Pinus* in the

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Box 1. Categorizing species sprouting behaviour

One of the obstacles to incorporating sprouting into plant strategy schemes^{a,b} is the difficulty of categorizing sprouting response. Sprouting varies among species, among life history stages of species and among disturbances of differing severity (Figs 1,2). A binary classification of species into sprouters and non-sprouters is widely used in ecosystems which experience regular crown fires such as Mediterranean type shrublands or boreal forests^c. However, even in these systems where severe disturbance is the norm, species range along a continuum from weakly to very strongly sprouting. Juvenile and adult sprouting ability often differs. In frequently burnt savannahs, sprouting is an essential prerequisite for juvenile

survival. However, if plants grow tall enough to escape fire damage, some species lose the capacity to sprout altogether although others remain vigorous sprouters. Forest trees show similar diversity in sprouting behaviour with many species sprouting as juveniles and losing the ability to sprout as adults^{d,e}.

Juvenile sprouting ability can be considered part of the recruitment strategy of a species. Adult sprouting behaviour, however, indicates potential persistence. It is a measure of whether a disturbance can cause tree or stand replacement. We therefore suggest that adult response to severe disturbance (causing death of most, perhaps all, of above-ground biomass) is the most useful stage to categorize species behaviour for

general comparisons. A binary classification into sprouter/nonsprouter is simplest given the current lack of information for most systems. However, these are really extremes of a poorly studied continuum.

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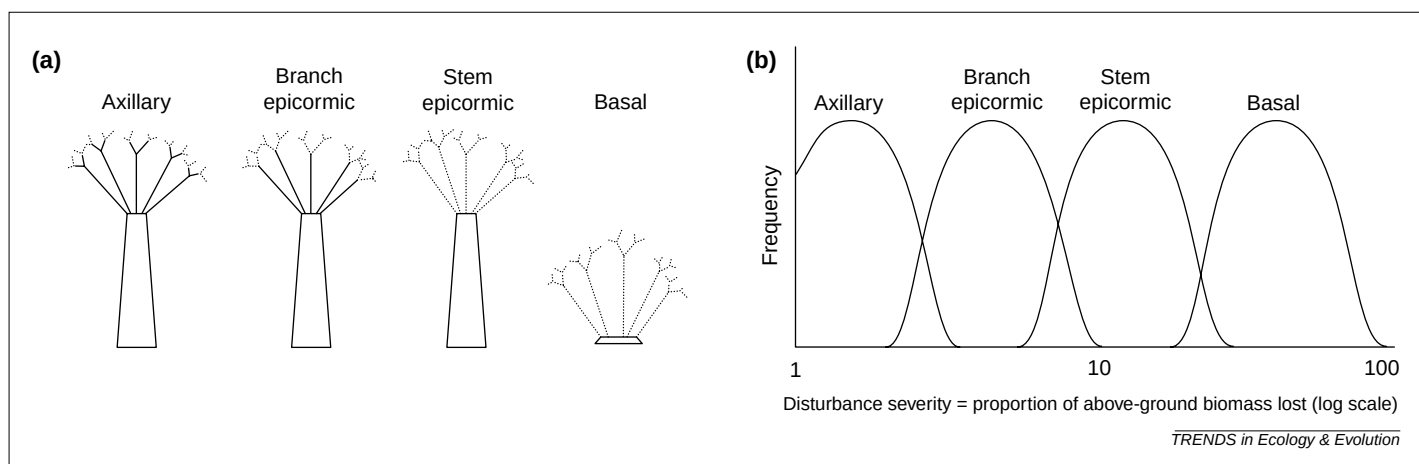


Fig. 1. The sprouting response varies with increasing disturbance severity. (a) Possible patterns of biomass loss and regeneration. Surviving biomass, unbroken lines; resprouted biomass, broken lines. (b) The relationship of sprouting response to disturbance severity (the proportion of above-ground biomass that is lost). *Reproduced, with permission, from Ref. 7.*

north and *Podocarpus* in the southern hemisphere⁵. Unfortunately, sprouting behaviour is rarely described in taxonomic monographs. In genera where it has been reported, sister species are sometimes difficult to distinguish on herbarium sheets but display striking differences in sprouting response¹³. Such species might be placed together in most plant strategy schemes^{2,3,14} but are ecologically distinct because of their different sprouting behaviour¹⁵.

Life history tradeoffs

To be able to sprout after sustaining an injury, a plant needs surviving meristems and stored reserves to support regrowth. Allocation of resources to storage carries a cost traded off against growth or reproduction^{16,17}. Indirect evidence for these tradeoffs has been documented in many studies in Mediterranean type shrublands^{4,8}. Sprouters generally have fewer seeds, smaller seedbanks, slower growth and maturation rates (from seeds), and almost always have fewer seedlings and poorer

seedling survival than nonsprouters in matched species comparisons⁴. Recently, similar tradeoffs have been reported for forest species. Seedling numbers decrease with increasing dominance of multi-stemmed sprouters in South African forests¹⁸, and a negative relationship between seedlings and sprouters has been reported for Jamaican forests recovering from cyclone damage¹⁹. The emerging generalization from both forests and fire-prone shrublands is that plants that sprout vigorously as adults tend to be poor recruiters, whereas nonsprouters recruit more readily.

Pate *et al.*²⁰ have explored the tradeoff more directly by studying carbon allocation patterns in a large sample of shrubs and graminoid species in Australian heathlands. On average, sprouters had root starch concentrations four times higher than those of nonsprouters, four to five times higher root:shoot ratios, and less than half the biomass of same-aged nonsprouters. Similar differences have been reported for shrubs of the Epacridaceae in Australia²¹ and

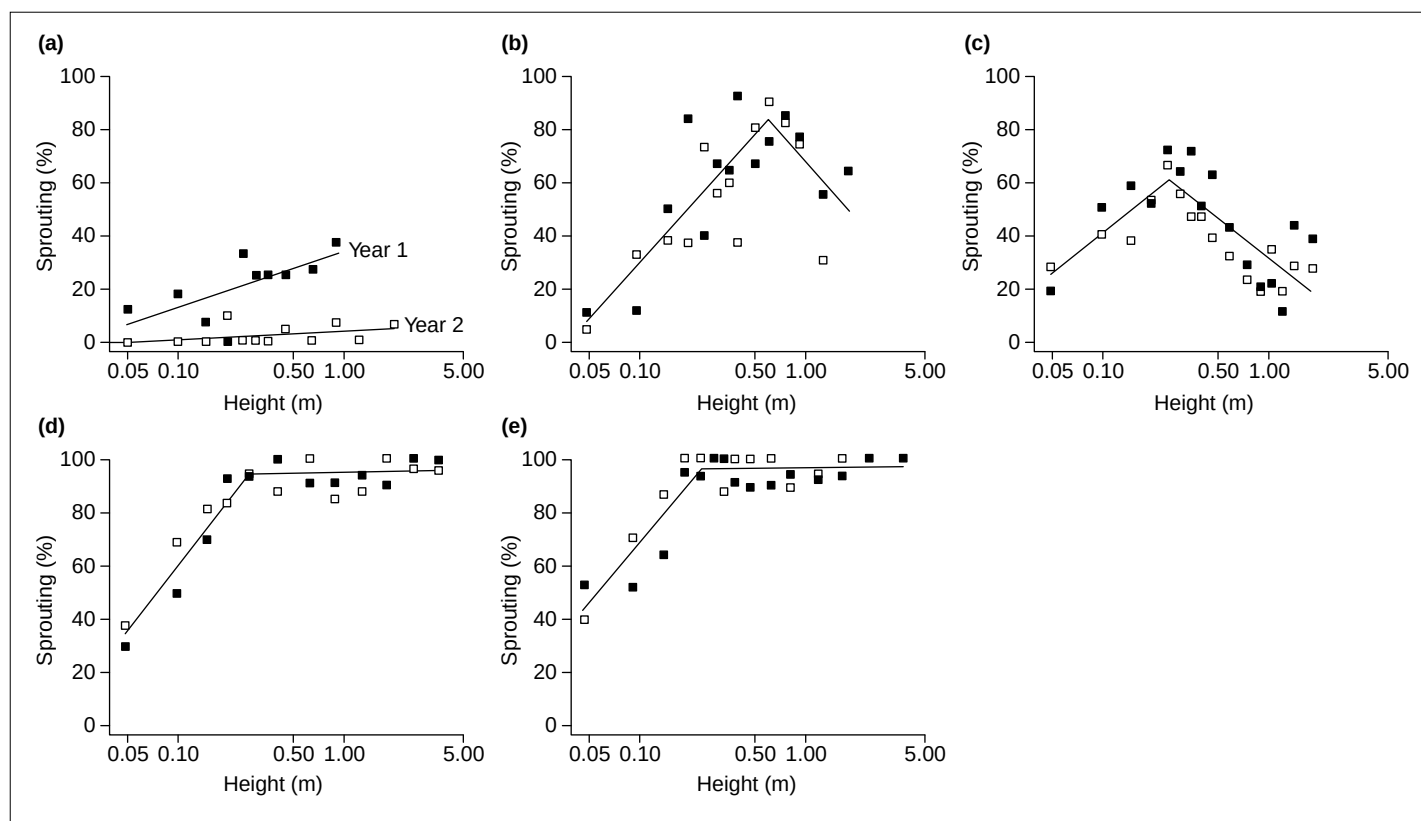


Fig. 2. The sprouting response to a disturbance of the same severity varies among species and with the size of the plant. Sprouting of shrubs (%) in different height classes of five Australian species subjected to a low intensity winter burn after two successive years (reproduced, with permission, from Ref. 9) *Acacia aneura* (a), *Dodonaea viscosa* (b), *Senna artemisioides* (c), *Eremophila mitchellii* (d) and *Eremophila sturtii* (e). Sprouting response can also vary with type of disturbance. The same species were cut, but not burnt, at the same time. Survival after 12 months was, for juveniles; *A. aneura* 35% for cut versus 28% for burned; *D. viscosa* 85% versus 65%; *E. mitchellii* 95% versus 93%; *E. sturtii* 95% versus 94%; *S. artemisioides* 75% versus 52%. For adults: *A. aneura* 25% for cut versus 0% for burned; *D. viscosa* 75% versus 52%; *E. mitchellii* 95% versus 90%; *E. sturtii* 95% versus 90%; *S. artemisioides* 95% versus 18%.

Ericaceae in South African fynbos²². Three *Erica* species show intraspecific variation with both sprouting and nonsprouting forms. Sprouters had root starch concentrations ranging from 5 to 35 times greater than nonsprouters, but starch concentration in shoots, which are burnt in fires, did not differ. The importance of carbohydrate reserves for resprouting has also been traced directly^{23,24}. Root reserves fell by 50–75% in *Stirlingia latifolia* (Proteaceae) when plants resprouted after a burn and took two years to recover to preburn levels²³. Repeated shoot removal in these studies caused a further reduction in root carbohydrates and eventual death of the plants.

As sprouters have lower biomass than nonsprouters of the same age, growth is a function of whole-plant allocation. The important implication is that plants with similar leaf properties might have widely divergent growth rates depending on the reserves allocated for resprouting. This point has also been made for saplings growing in temperate forests, where both data and models indicate that below-ground carbohydrate storage improves survival at the expense of growth, and is most favoured under low light conditions²⁵. Another implication is that species

that rely on stored carbon to subsidize resprouting could resprout more vigorously under elevated atmospheric CO₂ (Box 2).

We know of only one forest study, in the humid temperate forests of Japan, which explicitly compared carbon storage in sprouting and nonsprouting tree species²⁶. This study found abundant reserve carbohydrates in one of the frequently sprouting trees, *Quercus serrata*, but low reserves in another, *Euptelea polyandra* and in two species that rarely sprout. Resprouting of *Euptelea* apparently depends on the remobilization of above-ground resources and can only occur if some shoots survive damage from landslides to subsidize regrowth²⁷. Life history tradeoffs should be greater where stored reserves alone are used for re-sprouting than in species such as *Euptelea* which use photosynthetic subsidy.

Midgley²⁸ recently noted an additional cost of sprouting. In a variety of ecosystems, many sprouters are multi-stemmed and shorter than co-occurring nonsprouting relatives. He argued that multi-stemmed sprouters risk being shaded by single-stemmed, nonsprouting competitors and should therefore be restricted to less productive sites or areas with frequent disturbance^{7,18}. However, many sprouters are single stemmed, including tall forest species which retain a bank of sprouting shoots, and savannah trees which pass through a multi-stemmed juvenile phase before thinning to single stemmed adults. We also note that the tallest tree on earth, the Californian redwood (*Sequoia sempervirens*) resprouts after felling or fire damage and presumably eventually thins to single stems. Nevertheless, there are biomes

Box 2. Global change and the tree/grass balance

Resprouting rates are likely to be influenced by anthropogenic changes in atmospheric CO₂. Under elevated CO₂, both carbon storage and carbon allocation to stem sprouting is likely to increase. Because resprouting usually occurs in open disturbed environments rich in resources, CO₂ effects are likely to be greater than in closed communities. Bond and Midgley^a have recently suggested that elevated CO₂ impacts on woody resprouters could act as a general mechanism promoting woody plant invasion in fire-maintained grasslands and savannahs. Many mesic savannah trees need to reach a threshold size before juveniles escape topkill by fire. Under elevated CO₂, saplings are more likely to reach threshold size before another fire recurs (Fig. 1). The result could be a substantial increase in tree biomass in savannah environments with ecological repercussions of local and global significance^a. There is only one published study on elevated CO₂ effects on savannah tree resprouting. Hoffmann *et al.*^b simulated burning by clipping seedlings of *Keilmeyera coriacea*, a Brazilian savannah tree. Elevated CO₂ enhanced regrowth of clipped plants but only under high nutrient supply (such as might occur after a burn). Clipped plants accumulated greater root carbohydrate reserves under high CO₂ but the plants did not utilize these reserves for faster regrowth. Instead the seedlings appear to store carbohydrates in roots as insurance against future disturbance. The study used freshly germinated seedlings. Field experiments are needed to test whether elevated CO₂ might release saplings, trapped for decades by frequent burning, into the tree layer.

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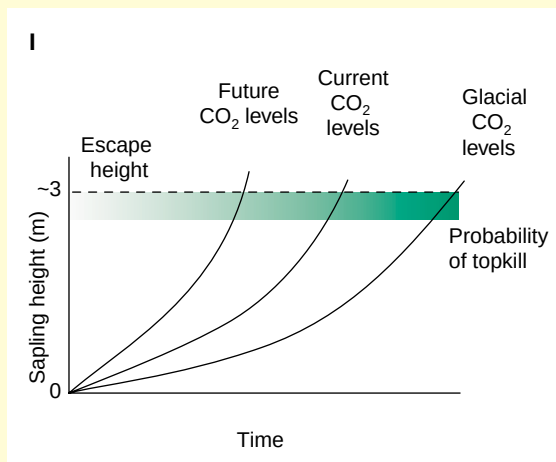


Fig. 1. The potential effect of different atmospheric CO₂ levels on resprouting of savannah tree saplings after fire (reproduced, with permission, from Ref. a). 'Probability of topkill' (represented by shading on the horizontal bar) increases with time as grass fuel accumulates, with a fire resetting plant height back to zero. Saplings can be held for decades within the influence of grass fires under current CO₂ conditions. Rising atmospheric CO₂ could accelerate sapling regrowth rates, thereby reducing the risk of topkill in the next fire and allowing saplings to recruit into the tree layer.

where sprouters are multi-stemmed and shorter than their single-stemmed, nonsprouting competitors^{7,18}.

Biogeography of sprouters

Patterns of sprouting in different regions are poorly documented. For fire-prone shrublands, Le Maitre and Midgley²⁹ tabulated data for Cape fynbos and noted that 52% of a sample of 4418 species were sprouters. Thus, even though conspicuous genera in the Cape are strongly dominated by nonsprouters

(e.g. *Protea*, *Leucadendron*), the flora is in general dominated by sprouters. In their review of data from forests recovering from windthrow, Everham and Brokaw⁵ found that the incidence of sprouting species is lower in temperate forest sites (mean 35.9%) than tropical sites (mean 51.5%) and that the number of sprouters tends to increase in more humid forests. The lower incidence of sprouting in temperate forest is partly due to the presence of conifers, as most adult trees in conifer forests are killed outright by severe disturbance⁵. Tropical forests not only have a high proportion of sprouting species but also a greater incidence of resprouting stems (mean 85.2%). A recent comparison of sprouting in logged and fire-degraded sites in several South American forests noted the highest levels of resprouting in areas disturbed by logging and fires³⁰. However, mature forests still contained significant proportions of continuously sprouting trees. For many other biomes, there is little information.

Sprouters seem to have the best of both worlds, possessing both vegetative and sexual reproduction. However, nonsprouters do occur and dominate in some genera. The question of what favours one or the other, or allows co-existence, has long troubled ecologists studying Mediterranean systems^{4,8,29}. One way of answering this is to consider biogeographic patterns. Lloret *et al.*³¹ looked at shrub responses to experimental fires in Mexican shrublands where fire is historically absent. Many of the species are congeners with Californian chaparral species although others have Neotropical affinities. If sprouting has evolved in response to burning in chaparral, one would expect fewer sprouters in Mexico. However, resprouting proved to be a widespread trait and many species had lignotubers and burls, sometimes interpreted as fire 'adaptations'. The near absence of nonsprouters in Mexico was the most conspicuous difference between these shrublands and Californian chaparral (where they are common). Selection for loss of sprouting is therefore the most significant feature of chaparral fire regimes and sprouting seems to be an ancient trait in these taxa.

At the scale of intragenetic patterns within ecosystems, two generalizations emerge. First, sprouter species tend to be more widely distributed (e.g. Lamont and Markey³² for *Banksia*) and second, sprouters tend to be more numerous in less productive sites^{7,28,33}. Why this should be the case requires further analysis. In productive areas, sprouters would be overtopped by single-stemmed seeders²⁸, but in unproductive habitats dependence on obligate reseedling is dangerous³³ because growth is slower and plants are more likely to be killed by disturbance without recruiting seedlings.

Another biogeographic perspective is linked to the type and intensity of disturbance. Frequent low intensity disturbance favours sprouters. Bellingham and Sparrow⁷ modelled the types of resprouters

versus the types of disturbance regimes. They suggest that basal resprouters should be most common in areas with severe and frequent disturbance, whereas axillary resprouters should be more common in areas with the least intense disturbance.

Role of sprouting in forest succession

Though sprouting is common in both temperate and tropical forests, and has long been important for traditional coppice management of woodlands, it has received very little attention in models and theories of forest succession or forest diversity. Instead the emphasis has been on recruitment limitations³⁴ or specialization of saplings for different microsite requirements such as those created by tree-fall gaps³⁵. However, recruitment limitation is only one side of the story; it ignores persistence. Long persistence of established trees reduces recruitment opportunities in both space and time, potentially reducing forest diversity. Sprouting will alter forest dynamics by favouring self-replacement after stem death, even if the tree is not shade tolerant. It is most important where stem death is caused by disturbance rather than crowding or disease. Does the sprouting habit confer long-term *in situ* persistence of forest trees? Does sprouting, rather than variation in seed supply or sapling growth requirements, determine local species composition in some forests? One way of exploring these questions is through the use of forest succession models such as SORTIE³⁶, which simulates forest succession by modelling crown competition for light and seedling recruitment limited by dispersal and light requirements. A variation of the SORTIE model has recently been used to test the idea that tree life history traits are adaptations to particular disturbance regimes³⁷. Loehle³⁷ asked whether species coexistence in forests could be predicted from the distribution of tree species in trait space under a given disturbance regime. Four independent dimensions were used to place species in 'strategy space': shade tolerance, adult tree size, root or stump sprouting ability, and seed dispersal distance. These are similar to Westoby's³ plant strategy scheme except that sprouting is included. Loehle notes that sprouting has 'not been considered previously as a major life-history attribute of trees or incorporated into plant demographic theory'. Simulations with a modified SORTIE model showed that the same species pool on a uniform substrate could produce different forest compositions depending on the disturbance regime. This result mirrors similar studies in the fire ecology literature^{4,15}. Loehle claims that the strategy space model can predict species richness in particular regions because species segregate into different tree niches with different combinations of the four traits. By combining possible positions on the trait axes, one can predict the number of species likely to occur in a given region.

However, for sprouting to be significant in forest succession, information is needed on the success of

sprouting as a means of persistence. Paciorek *et al.*³⁸ noted high levels of mortality of resprouts in long-term studies of a tropical forest on Barro Colorado Island. They suggest that for resprouting to be important, resprouters must recover reproductive status. We are not convinced that this statement is true (long-term persistence without recruitment will still influence forest dynamics). Nevertheless it is clear that if resprouting merely delays death by a few years it can be ignored in forest models. The availability of long-term data sets and the development of new forest models, promise better insights into the role of sprouting in forest dynamics in the near future.

Conservation biology

Sprouting ability is an important, but neglected, consideration for conservation and management of plant species^{39–41} (Box 3). Nonsprouters will be more vulnerable to recruitment failure after severe disturbance. They are more vulnerable to the problems associated with small population size, including inbreeding effects or loss of pollinators and dispersers³⁹. Sprouters are buffered from such effects. They are resistant to disturbance and catastrophe, can tolerate long periods with little or no recruitment and will tend to preserve genetic diversity, even in small populations, because of their long generation time. Populations of persistent sprouters would show little short- to medium-term response to the loss of pollinators and dispersers³⁹. The threat for sprouters is increased adult mortality caused, for example, by habitat alteration or by changing growing conditions. Poor recruitment compounds the problem because sprouters would be less likely to colonize unoccupied sites than nonsprouters. Rackham⁴² cites an interesting example from Groton Wood in Suffolk: *Tilia cordata*, a persistent sprouter, dominates ancient woodlands but still has not colonized an adjoining secondary woodland established in the 17th century.

Sprouting behaviour might also be important in determining tree species survival in the face of browsing by heavy mammals. In response to damage caused by elephants, for example, many African savannah tree species coppice vigorously but some, such as the baobab (*Adansonia digitata*), do not and thus suffer high mortality⁴³. Although we know of no comprehensive study of the problem, coppicing species should be better able to tolerate browsing damage than noncoppicing species. Monitoring the fate of nonsprouters, including their restriction to inaccessible cliff faces, could provide a sensitive indicator of environmental impacts of browsing by large mammals in national parks.

Evolutionary implications

Although it seems probable that differences in sprouting behaviour have important evolutionary consequences, there is a dearth of rigorous analysis.

Box 3. Conservation of sprouting versus nonsprouting *Banksia* species

The conservation implications of sprouting versus nonsprouting life histories are well illustrated by two studies of extinction risk for two rare Australian *Banksia* species^{a,b}. *Banksia cuneata*, a nonsprouter, only recruits after a fire, but is also killed by fire^a. Therefore, fires are essential for population growth; however, each fire risks local extinction because seedlings are vulnerable to periodic droughts that sometimes coincide with a burn. The drought risk can be avoided by not burning – but only temporarily. Plants live for only 50 years, thus long fire suppression will eventually cause extinction anyway. Conservation efforts should focus on successful recruitment after burns.

At the opposite end of the continuum, *Banksia goodii* is a vigorous rhizomatous sprouter – there are no records of adult mortality other than from habitat conversion. Seed production, let alone recruitment, has never been recorded in one third of populations. A recent analysis of extinction risk in this species predicted that, under the worst scenario, populations would decline by only 10% over 100 years and by 50% over 500 years^b! Clearly, conservation requirements for the two species are different. The main threats for the sprouter are loss of populations due to clearing of land for farming and roadways, and not recruitment failure. In general, persistent sprouters should require much less active management than nonsprouters unless adult mortality increases^c. Then, restoration or reintroduction is probably more difficult because vigorous sprouters tend to be poor recruiters.

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There are no comparative genetic studies of sprouters versus nonsprouters. Over 30 years ago, Wells¹² noted that nonsprouting species greatly outnumber sprouters in several chaparral genera. He suggested that differences in generation time influence speciation rates. Larger numbers of nonsprouting than sprouting species have been reported for many taxa in Mediterranean type shrublands²⁹. Selection for nonsprouting life histories might have contributed to diversification of these rich floras. However, greater diversity of nonsprouters is not universal and several genera have equal, or slightly higher, numbers of sprouting species^{13,32}.

Closely related sprouting and nonsprouting species occur in many genera in Mediterranean type floras and some species are polymorphic for the trait^{12,13,22}. This suggests that the evolutionary switch from sprouting to nonsprouting has occurred repeatedly. Explanations for selective pressures leading to these switches in life history have mostly been variants on how the tradeoffs between persistence and recruitment influence fitness in different environments^{4,7,8,22,31} or, possibly, benefits of shorter generation time for tracking environmental change¹². The evolutionary consequences of sprouting have still to be considered in forests, savannahs and other biomes.

Conclusions

Until recently, the literature on sprouting has been widely scattered with little attempt to explore ecological or evolutionary patterns. There are now several studies from widely divergent ecosystems that are taking a broader look at the phenomenon. To complement the regeneration niche¹, we suggest the use of the term 'persistence niche' for studies exploring ways in which established plants persist *in situ*. Sprouting behaviour is a key trait for persistence influencing the ecology of individuals, populations and communities. We suspect that the biology of persistence is at least as rich and varied as the biology of regeneration. Some of the questions of interest are:

- What is the distribution of sprouting in different ecosystems and different taxa?
- Why don't sprouters dominate the world, and why does their relative abundance vary in different communities and in different parts of the landscape?
- How general are the tradeoffs between sprouting and seedling recruitment?
- Does slow turnover in sprouter-dominated communities result in reduced species richness?
- Does sprouting influence competitive ability and reduce rates of successional change?
- Are there differences in the spread of sprouting and nonsprouting invasive species associated with recruitment tradeoffs?
- What are the evolutionary consequences of variation in sprouting behaviour?

We hope that a future review will be able to provide insights for a wider range of biomes and taxa.

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